CCCA Meta-Atlas — Griffiths et al. 2021 (Breast)

Breast Cancer (ER+ neoadjuvant) · 10x + iCell8 · 108,658 malignant cells analysed (67 samples ≥100)
Do malignant cells recapitulate embryonic programs? Non-branching (linear) manifold — GO / GO / NA

GO / NO-GO GATES

Gate 1 — mixing & continuity: GO

Harmony (θ=10) on sample integrates 67 samples across both technologies (10x+iCell8). Mixing judged on the per-neighbourhood iLISI (median ≈10.5/67), not the coarse same-sample scalar; continuity on the independent Harmony-PCA kNN graph (1 component). The manifold is one continuous, sample-mixed mass with a proliferative sub-region (not a confounded lineage); only ~1.4% detached specks removed → GO on the full manifold.

Gate 2 — module geography: GO (non-branching)

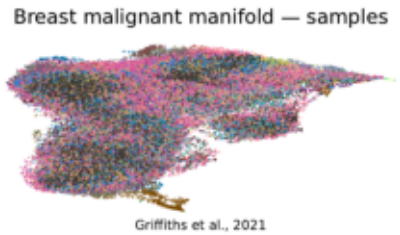
Organized cross-repression: smoothed pSom↔Crest r=-0.72 and Gast↔pSom r=-0.38 (raw pSom↔Crest -0.53). pSom and Crest occupy opposite territories on one continuous manifold — a single (Gast/pSom)→Crest inverse/terminal gradient, present but NOT resolved into two diverging arms → GO. No arm inversion to flag (Crest does not preferentially enrich a Gast arm).

Gate 3 — cross-repression: NA

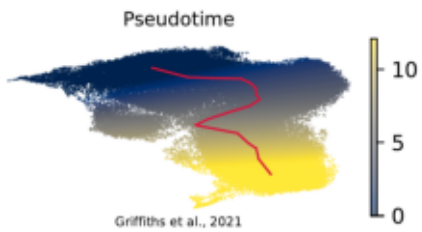
Non-branching topology (principal curve: 2 tips, 0 forks; fold-free self-approach 2.88). PPT "forks" grow with node count → space-filling, not a real bifurcation. No hub bifurcation to test → NA (branching-only correlation/modules notebooks not produced).

STEP FIGURES

Samples



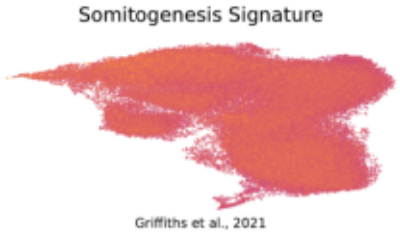
Pseudotime



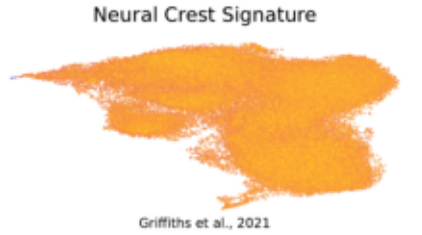
Gastrulation (raw)



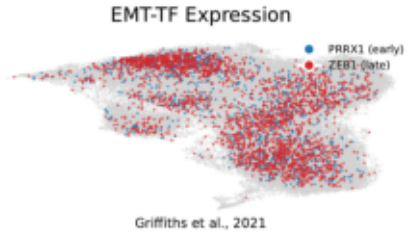
Somitogenesis (raw)



Neural Crest (raw)



EMT-TF Expression



PARAMETER CHOICES & RATIONALE

Threshold: ≥100 malignant cells/sample (re-derived). No dominant count cliff in the body of the distribution (median 646; only real gap in the 5→1 tail) → scale-appropriate floor guaranteeing a stable per-sample Harmony centroid at 10x scale. 67/95 samples, 108,658 of 111,372 malignant cells analysed.

Embedding: Harmony θ=10 on sample (strongest integration that converges; also removes the 10x/iCell8 batch, as samples nest within technology); n_pcs=30, n_neighbors=30, UMAP spread=1.3; 4,072 overdispersed genes; no sc.pp.scale before PCA.

Trajectory: scFates principal curve (non-branching), Nodes=20, epg_lambda=0.3, epg_mu=100; fold-free (self-approach 2.88).

Root: Gast/pSom-high, Crest-low curve tip (developmental origin; Crest is late/terminal). Axis not proliferation-driven (cycling vs pseudotime ρ=+0.03).

Geometry: Linear (single (Gast/pSom)→Crest axis). Raw module means: Gast -0.019, pSom -0.053, Crest +0.023.

EMT-TFs: PRRX1 (early) + ZEB1 (late), two most-detectable factors across families, detection-gated from a 10-TF panel incl. FOXC/PRRX (VIM excluded — a mesenchymal marker, not a TF). All EMT-TFs sparse (ZEB1 ≈9%, rest <6%) and pseudotime-flat.

Verdict: The embryonic pattern reproduces as a single gastrulation/somitogenesis→neural-crest axis (non-branching), with organized Gast/pSom↔Crest cross-repression (pSom↔Crest r=-0.72) and a terminal Crest gradient; no Gast/pSom bifurcation, so Gate 3 is NA. Geometry Linear — consistent with the other CCCA breast datasets. Resolves the prior science failure (gates never determined) to a clean GO / GO / NA call on the full continuous manifold (no lobe removal).